

ITZone

Entities, Attributes & Relationships
Database Design · Group 09

Entities & Attributes

1. CATEGORY

Attribute	Type	Note
category_id	INT	PK, Auto Increment
category_name	VARCHAR(100)	e.g. 'PC Components', 'Gear'
parent_category_id	INT	FK → category (self-ref, nullable)
description	VARCHAR(255)	Optional

2. CUSTOMER

Attribute	Type	Note
customer_id	INT	PK, Auto Increment
name	VARCHAR(100)	Full name
email	VARCHAR(150)	Unique
phone_number	VARCHAR(20)	Optional
address	VARCHAR(255)	Optional
password_hash	VARCHAR(255)	Hashed password
created_at	DATETIME	Auto set

3. PRODUCT

Attribute	Type	Note
product_id	INT	PK
name	VARCHAR(200)	
description	TEXT	
price	DECIMAL(10,2)	
stock	INT	
brand	VARCHAR(100)	
image_url	VARCHAR(500)	

Attribute	Type	Note
rating	DECIMAL(3,2)	Average rating
review_count	INT	
fps	INT	Performance metric
hz	INT	Performance metric
deal	BOOLEAN	Daily deal flag
category_id	INT	FK → category

4. PC_BUILD

Attribute	Type	Note
build_id	INT	PK
customer_id	INT	FK → customer
build_name	VARCHAR(150)	e.g. 'My Gaming Rig'
total_price	DECIMAL(10,2)	Auto-calculated
status	ENUM	draft / saved / in_cart / ordered
created_at	DATETIME	
updated_at	DATETIME	Auto updates on change

5. BUILD_COMPONENT

Attribute	Type	Note
component_id	INT	PK
build_id	INT	FK → pc_build
product_id	INT	FK → product
type	ENUM	CPU / GPU / Motherboard / RAM / SSD / HDD / PSU / Case / Cooling / Fan
quantity	INT	Supports 2x RAM, 3x Fans, etc.

6. COMPATIBILITY

Attribute	Type	Note
compat_id	INT	PK
product1_id	INT	FK → product
product2_id	INT	FK → product
is_compatible	BOOLEAN	TRUE = compatible
notes	VARCHAR(255)	e.g. 'Different socket type'

7. CART

Attribute	Type	Note
cart_id	INT	PK
customer_id	INT	FK → customer (UNIQUE — 1 cart per customer)
created_at	DATETIME	
updated_at	DATETIME	

8. CART_ITEM

Attribute	Type	Note
cart_item_id	INT	PK
cart_id	INT	FK → cart
item_type	ENUM	product / build
product_id	INT	FK → product (nullable)
build_id	INT	FK → pc_build (nullable)
quantity	INT	

9. ORDERS

Attribute	Type	Note
order_id	INT	PK
customer_id	INT	FK → customer
order_date	DATETIME	
total_amount	DECIMAL(10,2)	
status	ENUM	pending / confirmed / processing / shipped / delivered / cancelled
payment_method	ENUM	credit_card / KHQR
notes	TEXT	Optional delivery notes

10. ORDER_ITEM

Attribute	Type	Note
order_item_id	INT	PK
order_id	INT	FK → orders
item_type	ENUM	product / build
product_id	INT	FK → product (nullable)
build_id	INT	FK → pc_build (nullable)
quantity	INT	
price	DECIMAL(10,2)	Price snapshot at purchase time

11. PAYMENT

Attribute	Type	Note
payment_id	INT	PK
order_id	INT	FK → orders (UNIQUE)
method	ENUM	credit_card / KHQR
amount	DECIMAL(10,2)	
status	ENUM	pending / completed / failed / refunded
paid_at	DATETIME	Null until paid

12. SHIPPING

Attribute	Type	Note
shipping_id	INT	PK
order_id	INT	FK → orders (UNIQUE)
full_name	VARCHAR(150)	
phone	VARCHAR(20)	
address	VARCHAR(255)	
city	VARCHAR(100)	
country	VARCHAR(100)	Default: Cambodia
status	ENUM	pending / processing / shipped / delivered
shipped_at	DATETIME	
delivered_at	DATETIME	

13. PRODUCT_REVIEW

Attribute	Type	Note
review_id	INT	PK
product_id	INT	FK → product
customer_id	INT	FK → customer
rating	TINYINT	1–5 only
comment	TEXT	Optional
created_at	DATETIME	

Relationships

Relationship	Type	Description
CATEGORY → CATEGORY (self)	1 : M	A category can have many sub-categories
CUSTOMER → PC_BUILD	1 : M	One customer can save many builds
CUSTOMER → CART	1 : 1	One customer has one active cart
CUSTOMER → ORDERS	1 : M	One customer can place many orders
CUSTOMER → PRODUCT_REVIEW	1 : M	One customer can review many products
CATEGORY → PRODUCT	1 : M	One category has many products
PRODUCT → BUILD_COMPONENT	1 : M	One product can be used in many builds
PRODUCT <-> PRODUCT (via COMPATIBILITY)	M : M	Products can be compatible/incompatible with each other
PRODUCT → PRODUCT_REVIEW	1 : M	One product can have many reviews
PC_BUILD → BUILD_COMPONENT	1 : M	One build has many component slots
CART → CART_ITEM	1 : M	One cart holds many items
ORDERS → ORDER_ITEM	1 : M	One order has many line items
ORDERS → PAYMENT	1 : 1	One order has one payment record
ORDERS → SHIPPING	1 : 1	One order has one shipping record

Purchase Flow

#	Step	Detail
1	Customer browses	PRODUCT / GEAR catalog
2	Add individual product	CART_ITEM created (item_type = 'product')
3	Build custom PC	PC_BUILD created with status = 'draft'
4	Save build	PC_BUILD status → 'saved'
5	Add build to cart	CART_ITEM created (item_type = 'build'), PC_BUILD status → 'in_cart'
6	Review cart	CART + CART_ITEM displayed
7	Checkout	ORDERS record created, CART_ITEM → ORDER_ITEM (with price snapshot)
8	Payment	PAYMENT record created (pending → completed)
9	Shipping info	SHIPPING record created with delivery address
10	Order fulfilled	PC_BUILD status → 'ordered', ORDERS status → 'delivered'